Supplementary Information for
An attractor account of numerosity perception

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This document accompanies the article *An attractor account of numerosity perception*. Supplementary Material 1 details the trial-level fitting; Supplementary Material 2 reports the full model comparison; Supplementary Material 3 establishes trial-level parameter recovery; Supplementary Material 4 reports further model predictions and the individual-differences analysis; and Supplementary Material 5 presents the exploratory developmental-dyscalculia reanalysis.

Supplementary Material 1: Trial-level fitting

Data. We reconstruct one row per trial from the openly available data of Cheyette and Piantadosi (2020), grouping by session (`uniqueid`) and trial and keeping the presented numerosity, the typed response, and the response time (converted to seconds). The source paper analysed four between-experiment arms at $N = 100$ each after excluding the ten highest-error participants per arm; the open DURATION file pools the three duration-control arms (NONE / AREA / DENSITY) without that exclusion. Exposure duration is within participants: 379 sessions contribute DURATION trials and 360 completed all five exposures. We use the five exposure durations (40, 80, 160, 320, 640 ms), true numerosities $n = 1$–15, giving ≈10,600 trials per duration and 52,995 trials in total. No trials are aggregated: the likelihood is evaluated over the individual integer reports. Empirically, the recorded response time runs from stimulus offset (mask onset) to typed-response submission: the gap between consecutive trial timestamps equals the next trial's exposure plus its recorded RT plus a residual that is constant across durations (median 1101 ms), whereas measuring RT from stimulus onset would leave a residual that grows exactly with exposure. The exposure period is therefore excluded from the RT used here.

Sensory channel and well support. The completed sensory estimate is $y_0 \sim \mathcal{N}\big(n, \sigma(n)^2\big)$ with $\sigma(n) = \sqrt{s_{00}^2 + s_0\, n^{\lambda_0}}$ and fixed baseline $s_{00} = 0.05$; only s_0 and λ_0 are fitted (per duration in the joint fit). Integer memory components sit at $m = 1, \ldots, 19$ (the true range 1–15 plus four units of padding). The readout softmax and the energy V are defined over this same support. Typed responses run up to $R_{\max} = 99$; a few per cent fall outside the well support (3.9–6.1% across durations) and are handled by the lapse channel below.

Clean-up dynamics. From $x = y_0$ the state follows the discrete descent $x \leftarrow x - \mathrm{clip}(\eta\, \partial V/\partial x,\ \pm\tfrac{1}{2} \min_m \sigma_m)$ with fixed step size $\eta = 1$ and global clamp $\tfrac{1}{2} \min_m \sigma_m$ (equal to $\tfrac{1}{2}\sigma_c$ for the power-law wells). Descent stops at the first iteration t for which either $|\mathrm{step}| < \mathrm{tol}$ with $\mathrm{tol} = 10^{-4}$, or the step reverses sign relative to the previous update (oscillation criterion: $\mathrm{step}_t \cdot \mathrm{step}_{t-1} < 0$). In both cases the reversing or sub-threshold step is *not* taken, so the resting state x^* is the state at detection and the settling time is $T = t$. If neither criterion is met within the give-up budget G iterations, the trajectory is truncated: $T = G$ and x^* is the state after the G-th update. The RT links involving convergence time use this bounded step count T (main-text Eq. 3).

Give-up budget. On the flat large-numerosity landscape the descent does not converge, so without a bound the settling time—and with it the predicted response time—would grow without limit. We therefore stop clean-up after G steps. Profiling the joint response likelihood over $G \in \{40, 60, 100, 120, 140, 160, 250\}$, refitting the shared landscape (σ_c, p) and the per-duration sensory channel and lapse at each value, places the optimum near $G \approx 120$ ($\Delta \log L = 0$); $G = 100$ lies within about 5 log-likelihood units of this optimum ($\Delta \log L = -5.3$), whereas $G \le 60$ and $G \ge 250$ are disfavoured by tens to hundreds of units ($\Delta \log L = -86$ and -400). Across the plausible window $G \in [100, 160]$, the width index n_{width} varies by less than 0.2. We therefore use the profile-likelihood optimum, $G = 120$, and count it as one shared free parameter of every clean-up model. Because the budget shapes the large-n readout (mean estimate and exact-response rate), it is identified by the accuracy channel rather than by the response times, so using it to cap the settling time is not circular.

Response likelihood. For each model and duration the response probability of Eq. (4) marginalizes the readout over the sensory channel. The integral is computed *deterministically*, so the objective is free of Monte-Carlo

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Table S1. Fitted parameters of the joint winning model. Shared: $\sigma_c = 0.0185$, $p = 1.86$, $G = 120$, $s_{00} = 0.05$; selected RT link $\tau = 0.01144$, $\kappa = 0$, $t_0 = 1.30$ s (fixed). Sensory noise (s_0, λ_0), lapse ℓ, and RT residual σ_{rt} vary by duration (landscape held fixed for the RT fit). Well support $m = 1$–19; clean-up uses $\eta = 1$, $\mathrm{tol} = 10^{-4}$, and the oscillation stop above. We summarize the scale and exponent jointly through $n_{\mathrm{width}} = \sigma_c^{-1/p} = 8.5$.

Duration	s_0	λ_0	ℓ	σ_{rt}
40 ms	0.070	2.25	0.115	0.472
80 ms	0.0033	3.30	0.067	0.471
160 ms	0.0017	3.37	0.060	0.479
320 ms	0.00060	3.60	0.076	0.497
640 ms	0.00021	3.91	0.066	0.526

noise. Gauss–Hermite quadrature is inappropriate here: for the attractor models the clean-up map $y_0 \mapsto x^*$ is near-discontinuous (evidence snaps to basins), and Gauss–Hermite carries a 1.7–4% error even at 241 nodes. We therefore use a uniform standard-normal grid on $z \in [-7, 7]$ with Gaussian weights for *every* model, so the comparison is on identical footing. Against high-sample Monte-Carlo estimates, a grid of 801 nodes matches the integral to $\leq 2 \times 10^{-3}$ (mean 1.5×10^{-4}). The optimizer uses a cheaper grid (201 nodes); the reported log-likelihoods are recomputed at the fitted parameters on a fine grid (1601 nodes).

Lapse channel. We never discard out-of-support responses: the final response probability is $p_{\mathrm{final}}(r \mid n) = (1 - \ell)\, p(r \mid n) + \ell \,/\, R_{\max}$ with $R_{\max} = 99$, a uniform contamination over $r = 1, \ldots, 99$ that places positive mass everywhere. The lapse is counted in every AIC and BIC. It is fitted separately at each duration in the attractor, static-readout, Weber/ANS, and C&P fits. The published-form joint C&P fit instead shares one lapse across durations; we therefore also fit a matched joint C&P variant with duration-specific lapses.

Response-time channel. Response time is modelled as log-normal, $\log \mathrm{rt} \sim \mathcal{N}(\log(t_0 + m(n)), \sigma_{\mathrm{rt}})$, so main-text Eq. (3) is the location of this log-normal (the predicted mean RT). The metric $m(n)$ is the expected posterior entropy ($\kappa\, \mathbb{E}[H]$), the expected convergence time ($\tau\, \mathbb{E}[T]$), or their sum (the hybrid). We fix the basal latency at $t_0 = 1.30$ s: because RT is timed from stimulus offset, this intercept absorbs mask-viewing, typing, and motor time rather than exposure duration, and at the longest exposures the geometric-mean RT for $n = 1$ approaches $\sim$1.30 s, placing t_0 at the floor of the observed chronometry. It is held fixed for all models and durations and is not estimated from the likelihood. Link coefficients are shared across durations and residual scatter σ_{rt} is profiled per duration. The steps-only link is selected by AIC, narrowly ahead of the hybrid ($\Delta\mathrm{AIC} = 0.24$) and far ahead of entropy alone ($\Delta\mathrm{AIC} = 1579.5$). All three are fitted with the landscape held fixed at its response-fit values, providing a constrained cross-channel test of the fitted landscape.

Optimization. The deterministic objective is minimized by `scipy` differential evolution (Storn and Price 1997) followed by a Nelder–Mead polish, with fixed seeds.

Supplementary Material 2: Model comparison and constrained fits

Table S2 shows the joint response-model fits and states the sharing configuration used for every row. In the power-law attractor and static fits, only the landscape parameters (σ_c, p) are shared; in the constant-width ablations, $p = 0$ is fixed and only σ_c is shared. The sensory parameters (s_0, λ_0) and lapse are duration-specific. The Weber/ANS model has no separable landscape, so its four parameters remain duration-specific.

The normative benchmark is the bounded-optimal model of Cheyette and Piantadosi (2020), reimplemented from their published code: a Zipf $(1/n^2)$ prior read out through an information-bottleneck posterior whose information budget is $\min(B, R\,t)$ bits, with t the exposure duration. Its published joint form shares B, R, and one lapse across durations ($k = 3$). Because the other joint models use duration-specific lapses, the matched C&P comparison shares only B and R and fits five lapses ($k = 7$); this is the C&P row used in main-text Fig. 2a. Our implementation reproduces the published parameters on simulated data and matches the reference likelihood to machine precision.

Turning the clean-up dynamics on adds a single parameter, the shared give-up budget G (Supplementary Material 1); the dynamics are otherwise deterministic given the landscape. Even net of this penalty, the attractor improves AIC by 721.6 over its matched static readout in the joint fit. Fixing $p = 0$, and hence $\sigma_n = \sigma_c$, worsens AIC by 3488.2 (both models share the budget, so this comparison is unaffected). Clean-up therefore remains beneficial under the ablation, but it cannot compensate for removing magnitude-dependent broadening.

A further alternative dispenses with the completed estimate altogether. In this purely driven model there is no sensory channel $\sigma(n)$: the items are registered one at a time over the exposure and the running count itself drives the attractor, while the landscape pulls the state towards the stored integers (main-text Methods), with the response read out at offset. It fits notably worse than the completed-estimate attractor (Table S2). The reason is visible in the trajectories (Fig. S1): on the sharp small-number wells the state is captured by whatever integer the partial count currently resembles and is dragged up through the wrong wells as the count grows, because a partial count is not diagnostic of the total. A partial count is thus a poor retrieval cue, unlike a degraded but complete estimate.

The width exponent p is a free parameter of the model, shared across durations and fitted at $p = 1.86$ (Table S1). It grows the wells from narrow and well-separated at small n to wide and overlapping at large n, but p and the scale σ_c are not interpreted singly. We therefore summarize their joint effect with the continuous width-based index n_{width}, defined by $\sigma_n = 1$. What separates our account from the Weber/ANS magnitude code is not a particular width exponent or this proxy, but the discrete pattern-completion architecture: at the identical trial-level convention the attractor beats the magnitude code by $\Delta\mathrm{AIC} = 838$ (Table S2). Fitted sensory, lapse, and RT-link values for the joint winning model are reported in Table S1 (Supplementary Material 1).

The comparison shares one landscape across durations. The attractor beats the Weber/ANS fit by $\Delta\mathrm{AIC} = 838$

Table S2. Trial-level response-model comparison (joint shared-landscape fit). k is the total number of free parameters over all five durations. Information criteria use all $52{,}995$ responses and are reported relative to the power-law attractor. The joint power-law models share σ_c and p; the constant-width ablations fix $p = 0$ and share only σ_c. In all cases, s_0, λ_0, and lapse are fitted per duration. The clean-up (attractor) models additionally count the shared give-up budget G (Supplementary Material 1) in k, so turning the clean-up on adds one parameter. Weber/ANS remains fully duration-specific. The C&P model shares B and R but uses a duration-specific lapse ($k = 7$).

Model	k	ΔAIC / ΔBIC
Power-law attractor	**18**	**0 / 0**
Driven (accumulated count)	11	7954 / 7891
Power-law readout (static)	17	722 / 713
Constant-width attractor ($p = 0$)	17	3488 / 3479
Constant-width readout (static; $p = 0$)	16	3718 / 3700
Weber/ANS	20	838 / 855
Normative (C&P 2020)	7	11453 / 11355

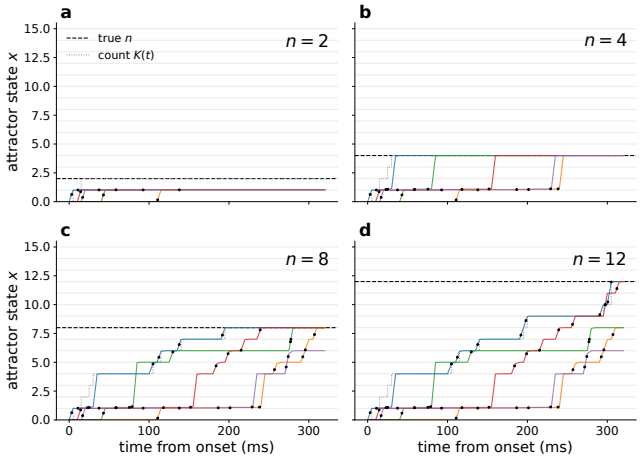


Figure S1. The purely driven model captures the wrong integer. State $x(t)$ of the purely driven model on the winning (sharp) landscape at 320 ms, for five example trials each at **a**, $n = 2$; **b**, $n = 4$; **c**, $n = 8$; **d**, $n = 12$. Black dots mark item registrations; the dotted line is one trial's running count $K(t)$; the dashed line is the true n. Driven by the accumulating count, the state is captured by whichever integer the partial count currently resembles and is dragged up through the wrong wells; for small n it is trapped near 1, and for large n it has not reached n by offset, with high across-trial variance. A partial count is not diagnostic of the total, so it is a poor retrieval cue.

(ΔBIC $= 855$) and the matched C&P fit by ΔAIC $= 11{,}453$ (ΔBIC $= 11{,}355$).

Supplementary Material 3: Parameter recovery

To confirm that the fitted quantities are identifiable at the trial level, we ran a parametric recovery: for each duration we generated 20 synthetic datasets from parameters jittered around the fitted solution, sampled with the empirical per-numerosity trial counts and the lapse channel, and refitted them with the exact headline pipeline (100 refits in total). Table S3 reports the recovery of each parameter and of the derived quantities the paper interprets; Figure S2 shows the scatter and the error-correlation structure.

The landscape scale σ_c and exponent p trade off strongly (error correlation $\rho = -0.97$) and are not identifiable on their own, as do the two sensory parameters s_0 and λ_0 ($\rho = -0.98$). The data constrain the combined width profile rather

Table S3. Trial-level parameter recovery of the winning model. Identity R^2 and Spearman ρ between generating and recovered values across the 100 synthetic datasets. Scale parameters are compared on the log scale. The interpreted quantities are the derived ones (lower block): n_{width} is a heavy-tailed transform of the strongly traded-off (σ_c, p), so its identity R^2 is uninformative and it is summarised by rank.

Quantity	R^2	Spearman ρ
σ_c (well scale)	0.66	0.76
p (width exponent)	0.15	0.81
s_0 (sensory scale)	0.82	0.94
λ_0 (sensory exponent)	0.41	0.83
ℓ (lapse)	0.96	0.98
n_{width} (derived)	—	0.98
σ_3 (well width at $n = 3$)	0.89	0.89
$\sigma(n = 5)$ (sensory sd)	0.99	1.00

than its scale and slope independently. We therefore interpret the combined sensory-noise profile $\sigma(n) = \sqrt{s_{00}^2 + s_0 n^{\lambda_0}}$ ($\sigma(n = 5)$, $R^2 = 0.99$), the fitted width σ_3 ($R^2 = 0.89$), the lapse ($R^2 = 0.96$), and the width index n_{width}, which recovers in rank (Spearman $\rho = 0.98$, median error 0.04 numerosities). Since $n_{\text{width}} = \sigma_c^{-1/p}$ is a heavy-tailed transform of the traded-off pair, a handful of ridge fits admit a low-exponent solution of near-equal likelihood that inflates the pointwise estimate. The index is robust for ordinal comparisons of the fitted width profile, but it is not an estimate of where attractors disappear.

Supplementary Material 4: Model predictions and individual differences

Exact-accuracy shoulder. The predicted proportion of exact responses shows a small shoulder near n_{width}, even though the fitted width law and sensory-noise profile vary smoothly with n (main-text Fig. 3b). The shoulder vanishes when clean-up is switched off at identical parameters (Fig. S3), so it reflects the dynamics on the integer lattice. Its numerical alignment with n_{width} does not make the width proxy a discrete exact-range boundary.

Individual differences (main-text Fig. 2c). We also ask whether a sharper landscape goes with lower estimation noise across participants, using a bivariate hierarchical model that avoids overlapping summary statistics. Clustering is by session, and we retain every typed response at 320 and

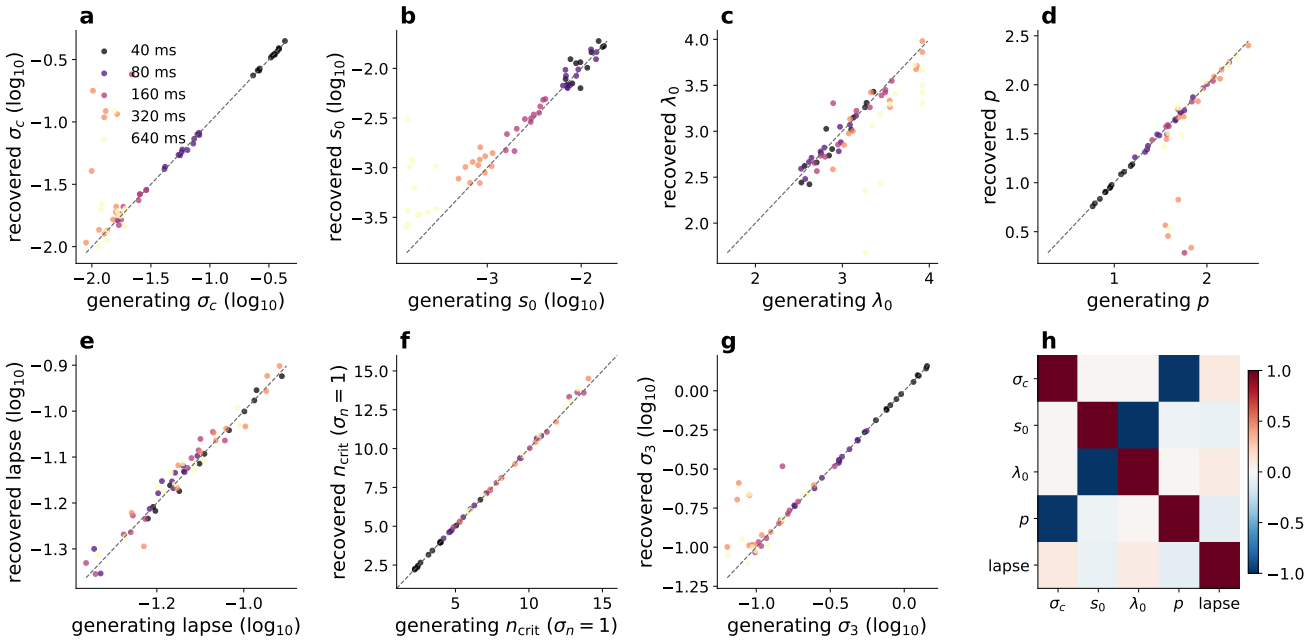


Figure S2. Parameter recovery of the winning model. Recovered versus generating values for the five fitted parameters and the derived n_{width} and σ_3, coloured by exposure duration; dashed lines are identity. The n_{width} panel is clipped to its meaningful range: the on-diagonal bulk recovers well, while a few ridge fits (low exponent, near-equal likelihood) fall off-scale. The bottom-right panel is the Spearman correlation matrix of recovery errors, showing the two strong trade-offs, σ_c–p and s_0–λ_0.

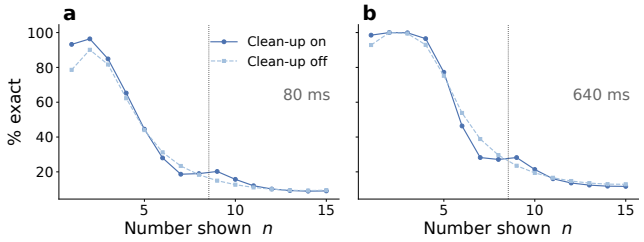


Figure S3. Clean-up generates an exact-accuracy shoulder. The shoulder is generated by the integer clean-up dynamics and disappears in the matched static control. It is more pronounced than in the behavioural data, making it both a distinctive prediction of the mechanism and a residual of the current homogeneous landscape. Predicted proportion of exact responses at 80 ms (**a**) and 640 ms (**b**) with clean-up on (solid) and off (dashed), at identical parameters; dotted verticals mark n_{width}. The shoulder occurs near this width-based proxy and disappears without clean-up.

640 ms for the 366 sessions that completed both exposures, and require at least one trial in each channel of a given split ($N = 362$). Small-n trials ($n \le n_{\text{lo}}$) enter a power-law attractor categorical likelihood with participant random effect u_i^n on $\log n_{\text{width}}$ and duration-specific fixed effects; the width exponent p and the sensory channel (s_0, λ_0) are held at the group joint-winner values, and the duration-specific lapses use the main $R_{\max} = 99$ convention. Large-n trials ($n \ge n_{\text{hi}}$) enter a discretized linear Weber model $r^\star \sim \mathcal{N}(\gamma_d\, n, (w\, n)^2)$ on $r = 1, \ldots, 99$, with participant random effect u_i^w on $\log w$, duration-specific fixed effects, and a pooled duration-specific gain γ_d. The latent pair (u_i^n, u_i^w) is bivariate normal with correlation ρ (partial pooling via Gauss–Hermite marginalization). The primary split is $n_{\text{lo}} = 8$, $n_{\text{hi}} = 9$; neighbouring splits $7/8$ and $9/10$ are reported as robustness checks. At the primary split, $\hat{\rho} = -.569$, likelihood-ratio $\chi^2(1) = 156.9$, $p < 10^{-35}$. The neighbouring splits give $\hat{\rho} = -0.767$ and -0.709. Main-text Fig. 2c plots empirical-Bayes n_{width} against w (duration-averaged) under the primary split.

Supplementary Material 5: Exploratory reanalysis of developmental dyscalculia

As an exploratory extension we applied the winning model to the exact-enumeration data of Decarli et al. (2020): accuracy and response time per subject and numerosity for 32 children with developmental dyscalculia (DD) and 32 age-matched controls, using numerosities $n = 1$–7 (dropping $n = 8$ exactly as the original authors do, for the guessing end-effect they report). The dot arrays were shown for a fixed exposure of 500 ms, so all quantities in this section refer to that exposure. The model fits both groups well (Fig. S4a,b) and again reproduces the widening well profile of the main dataset. We summarize each child's fitted width profile with the index n_{width}, defined as in the main text.

First, the decline of accuracy with numerosity is better explained by changing the fitted well-width profile than by sensory noise. With an exhaustive per-subject grid search and a fixed common lapse and exponent, we fitted two single-mechanism variants to each child's accuracy curve. In the width-only variant, sensory noise is switched off and the decline is carried by broader, more overlapping wells. In the noise-only variant, the width profile is fixed and sensory noise carries the decline. The width-only variant fits better in $64/64$ subjects (mean SSE 0.010 vs. 0.097, Wilcoxon $p = 3.5 \times 10^{-12}$; Fig. S4c). Within these two restricted single-mechanism variants, the width-only account fits the individual accuracy curves better than the noise-only account.

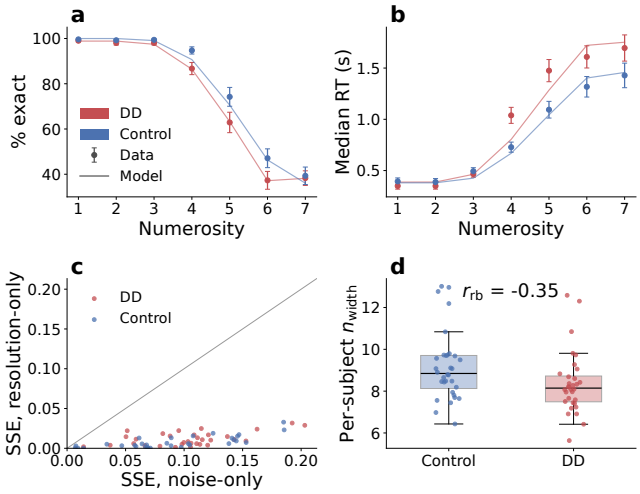


Figure S4. Developmental dyscalculia and a coarser fitted width profile. (a) Exact accuracy per numerosity (points, group mean ± SEM; lines, the group-level model fit). (b) Median response time, same conventions. (c) Per-subject accuracy SSE of the two single-mechanism fits: the width-only model fits better than the noise-only model in $64/64$ subjects (all points below the diagonal). (d) Per-subject width index n_{width} from free-scale fits with a shared exponent at the task's 500 ms exposure. Boxes show the group interquartile range and median; children with DD have lower n_{width} than controls (Mann–Whitney $p = 0.018$, $r_{\mathrm{rb}} = -0.35$).

Second, the group difference is reflected in the fitted width profile. Children with DD have lower extrapolated n_{width} than age-matched controls (median 8.1 vs. 8.8, Mann–Whitney $p = 0.018$, $r_{\mathrm{rb}} = -0.35$), whereas the mirror analysis that frees sensory noise instead shows only a weak trend ($p = 0.05$) and fits worse. In rank terms, the effect is comparable to the raw accuracy deficit at $n = 4$–6 ($r_{\mathrm{rb}} = -0.33$). The result is consistent with a coarser fitted numerical code in DD, although the analysis remains exploratory.

On this model, then, developmental dyscalculia is associated with a coarser fitted numerical code. It is also complementary to the account of Dolfi et al. (2024), who locate the dyscalculic deficit in a weaker number sense within an image-computable network. Both accounts place the deficit in the number code, and coupling the attractor to such a front end (which the present model sets aside) would allow a resolution deficit and a sensory-noise deficit to be separated within a single model. The analysis remains exploratory: the data are coarse (7 trials per cell), the localization to width profile rests on the per-subject model comparison rather than on a full identification of the two mechanisms, and a small sensory-noise component may coexist.

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